Phase-assignment controls: numerical summary

A. branching genes

		normal + branching (n=20)		hybrid: branching (n=20)	
Method	Condition	$CBDir_{gene}$	CosSim	$CBDir_{gene}$	CosSim
scVelo_dyn	Original	0.3630	0.6118	0.4454	0.6096
scVelo_dyn	Induction only	0.4927	0.8008	0.5585	0.7954
MultiVelo (RNA only)	Default	0.5003	0.8205	0.5663	0.8143
MultiVelo (RNA only)	Gate open	0.2054	0.2477	0.2528	0.2349
MultiVelo (RNA only)	No output smoothing	0.5007	0.8218	0.5664	0.8151
MultiVelo (RNA only)	Complete trajectory	0.3070	0.5361	0.3995	0.5403

B. MURK genes

		normal + MURK (n=10)		hybrid: MURK (n=20)	
Method	Condition	$CBDir_{gene}$	CosSim	$CBDir_{gene}$	CosSim
scVelo_dyn	Original	0.5884	0.7114	0.4230	0.5177
scVelo_dyn	Induction only	0.7223	0.8219	0.4618	0.5656
MultiVelo (RNA only)	Default	0.6591	0.7443	0.4989	0.5995
MultiVelo (RNA only)	No output smoothing	0.6986	0.8068	0.4989	0.5980
MultiVelo (RNA only)	Complete trajectory	0.6323	0.7114	0.3732	0.4621

C. Gate-open paired checks (number of genes with lower values)

Subset	$CBDir_{gene}$	CosSim	Fitting loss
normal + branching	20/20	20/20	20/20
hybrid: branching	20/20	20/20	20/20